Numerical Methods and Optimization

16/2/2026 – 3^{rd} Year

ENSIA — Spring 2026

Tutorial 2

- Approximate resolution of nonlinear equations of type $f(x) = 0$ -

Exercise 1: Roots separation

Separate the roots of the following equations using a graphical method (use a calculator or *plot* in Octave), algebraic method (factorize the polynomials), or analytical method (study the variations of the function). Provide the multiplicity of the roots if possible.

(1) $\left(u^3 - 6u^2 + 9u\right)\left(e^u sin(u) - 1\right) = 0$ in $]-\pi, \pi[$ (2) $tanz = z$ in $\mathbb{R}_+^*$

Exercise 2: Bisection Method

(1) Calculate $\sqrt{2}$ with a calculator equipped only with the four basic operations.

(2) A binary classifier outputs a probability $p(x)$ that a sample belongs to class 1. The classifier is trained on a dataset, and its overall accuracy depends on the **decision threshold** T: if $p(x) \geq T$, the sample is classified as 1; otherwise, it is classified as 0.

We want to find the threshold T that maximizes the **F1-score** of the classifier. The relationship between the F1-score and the threshold T is approximated by the function:

$$F(T) = \frac{2 \cdot p(T) \cdot r(T)}{p(T) + r(T)} - F_{\text{target}} = 0$$

where $p(T)$ and $r(T)$ are the precision and recall functions of the model for threshold T, and $F_{\text{target}} = 0.80$ is the target score. The following values are given:

$$p(T) = 0.5 - 0.5\sqrt{T}, \quad r(T) = 0.5 + 0.5T^2$$

a. Use the **bisection method** to perform **3 iterations** and obtain an approximation of the threshold T that reaches the target score F_{target}.

b. How many iterations would be required to achieve a 1% precision on the threshold T?

Exercise 3: Method of Newton-Raphson

Suppose it costs $C(p)$ dollars to produce p grams per day of a certain chemical sach that:

$$C(p) = 2p^{\frac{1}{3}} + 3p + 200$$

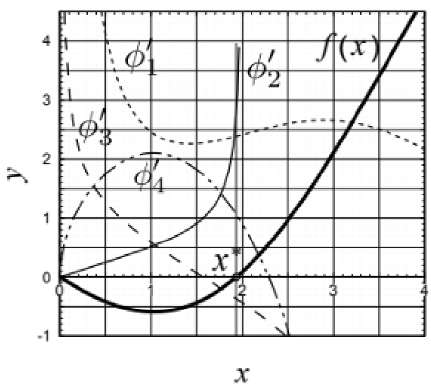
The firm can sell any amount of the chemical at 4$ a gram. Find the break-even point of the firm, that is, how much it should produce per day in order to have neither profit nor a loss (use with an initial guess as $p_0 = 250$ and stop when error reaches 0.1%).

Exercise 4: $f(x) = 0$ to fixed point method

The equation $f(x) = \frac{x^2}{4} - \sin x = 0$ has a unique positive root x^*

Verify that $f(x) = 0$ is equivalent to $x = \varphi(x)$, then use the figure below to determine which of the four functions φ_i can be used to determine x^* by the method of successive iterations $x_{n+1} = \varphi_{(x_n)}$

1. $\phi_1(x) = x + \frac{x^2}{4} - \sin x$
2. $\phi_2(x) = \sin^{-1}\left(\frac{x^2}{4}\right)$
3. $\phi_3(x) = 2\sqrt{\sin x}$
4. $\phi_4(x) = 2\sqrt{\frac{3x^2}{4}\sin x}$



Exercise 5: Fixed point Method

In a simple iterative optimization algorithm (like gradient descent), the learning rate α is adjusted to reach a desired convergence speed. Suppose the update rule for α is approximated by the function:

$$\alpha_{n+1} = g(\alpha_n) = 0.6 + 0.25\ \ln(1 + \alpha)$$

where α_n is the learning rate at iteration n. We want to find the fixed point α^* such that:

$$\alpha^* = g(\alpha^*)$$

(1) Discuss the convergence of the method. does the iteration converge quickly or slowly, and why?

(2) Use the **fixed point iteration method** to perform **5 iterations** starting from $\alpha_0 = 0.6$ and obtain an approximation of the fixed point α^*.

(3) How many iterations are required to guarantee a precision of 0.1% on the learning rate?

(4) Does the iteration converge quickly or slowly, and why?

(5) Explain why finding a stable learning rate is important in training machine learning models.